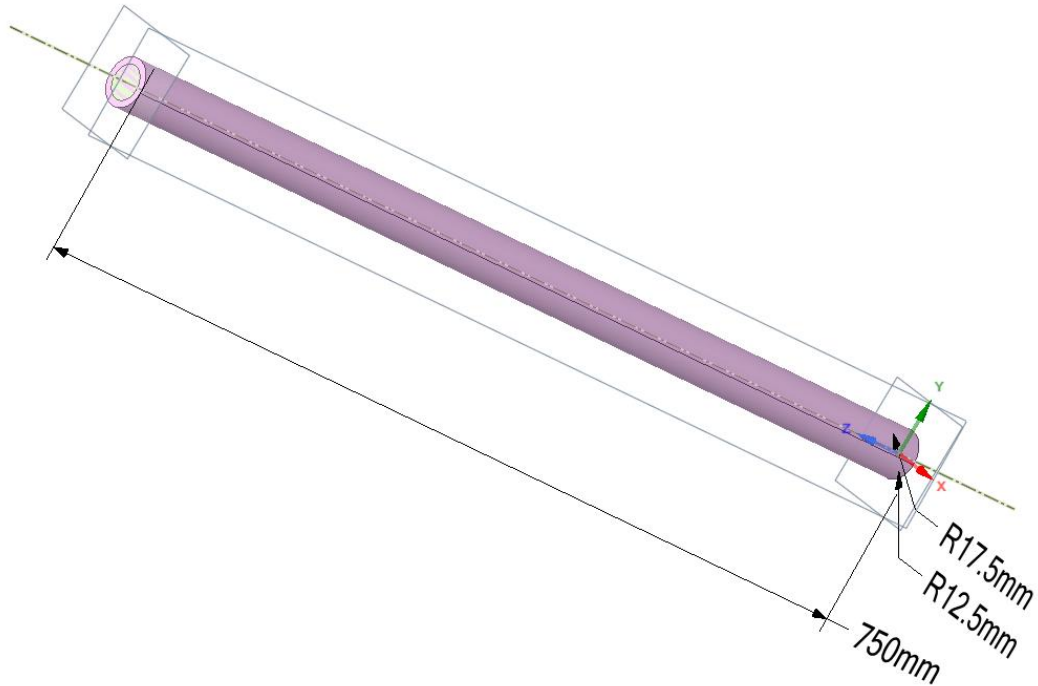


Composite Pillar Structural Analysis

Aim : To determine the deformation and stress of the composite pillar

Geometry:

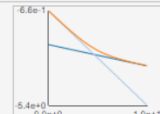
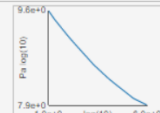


Outer Diameter-17.5mm

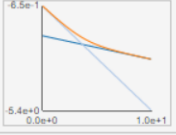
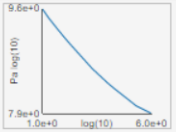
Inner Diameter-12.5mm

Length-750mm

Material Selection : For the Inner rod- Structural Steel

Structural Steel	
Fatigue Data at zero mean stress comes from 1998 ASME BPV Code, Section 8, Div 2, Table S-110.1	
Density	7850 kg/m ³
Structural	
Isotropic Elasticity	
Derive from	Young's Modulus and Poisson's Ratio
Young's Modulus	2.1e+11 Pa
Poisson's Ratio	0.3
Bulk Modulus	1.75e+11 Pa
Shear Modulus	8.0769e+10 Pa
Isotropic Secant Coefficient of Thermal Expansion	1.2e-05 1/°C
Compressive Ultimate Strength	0 Pa
Compressive Yield Strength	2.5e+08 Pa
Strain-Life Parameters	
S-N Curve	

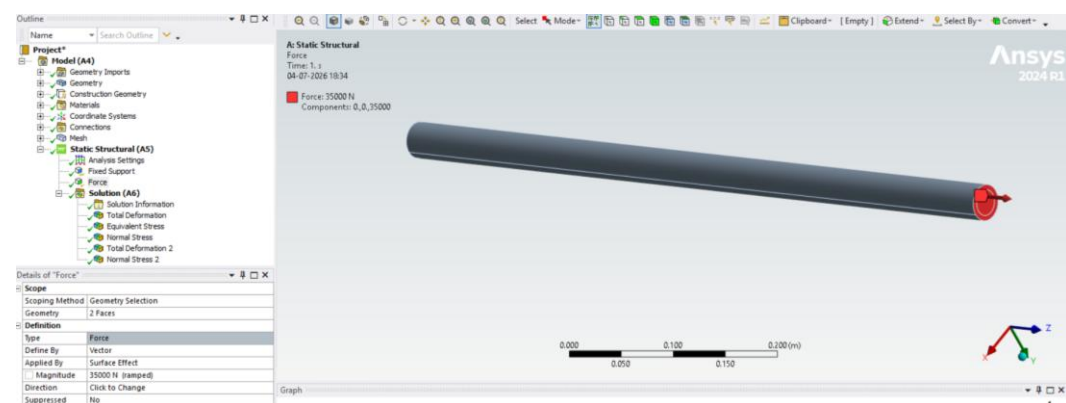
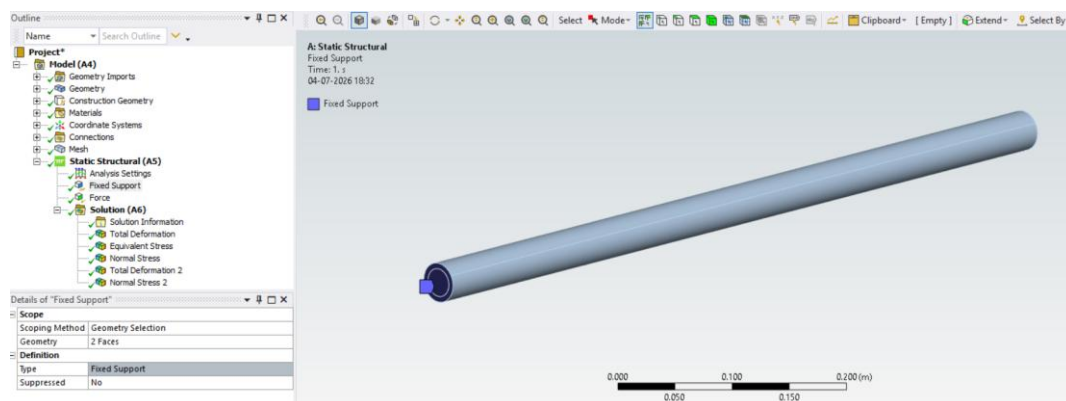
Material Selection : For the Outer rod- Brass

Brass	
Fatigue Data at zero mean stress comes from 1998 ASME BPV Code, Section 8, Div 2, Table 5-110.1	
Density	8700 kg/m ³
Structural	
Isotropic Elasticity	
Derive from	Young's Modulus and Poisson's Ratio
Young's Modulus	9.5e+10 Pa
Poisson's Ratio	0.3
Bulk Modulus	7.9167e+10 Pa
Shear Modulus	3.6538e+10 Pa
Isotropic Secant Coefficient of Thermal Expansion	1.2e-05 1/°C
Compressive Ultimate Strength	0 Pa
Compressive Yield Strength	2.5e+08 Pa
Strain-Life Parameters	
S-N Curve	

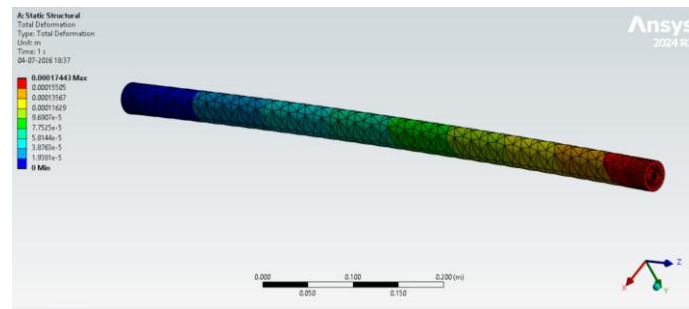
Boundary Condition:

One end is fixed

Another end is with the force- 35KN



Result : Total Deformation



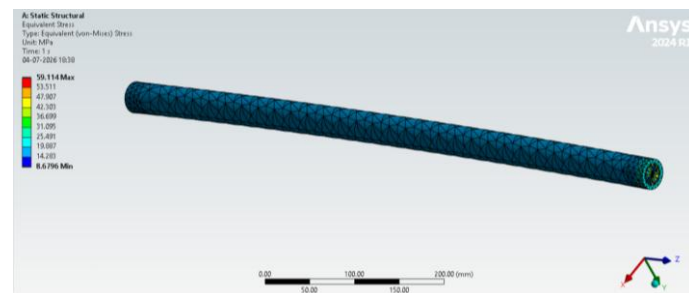
The total deformation result illustrates the displacement of the composite pillar under the applied static loading condition. The pillar experiences a maximum deformation of approximately 0.000174 m (0.174 mm) at the free end, while the fixed end exhibits zero displacement, satisfying the prescribed boundary conditions.

The deformation gradually increases along the length of the pillar, forming a smooth displacement gradient from the fixed support to the loaded end. This behavior is characteristic of a cantilever-type structural member subjected to an axial or transverse load. The relatively small deformation indicates that the composite pillar possesses high stiffness and maintains its structural integrity under the applied loading.

Observation

- Maximum Total Deformation: 0.174 mm
- Minimum Total Deformation: 0 mm
- Maximum deformation occurs at the free end.
- Deformation varies smoothly along the pillar length, indicating stable structural behavior.

Result : Equivalent Stress



The equivalent (Von Mises) stress distribution was evaluated to determine the stress concentration within the composite pillar. Figure 4.2 shows that the maximum equivalent stress is approximately 59.114 MPa, while the minimum stress is 8.676 MPa.

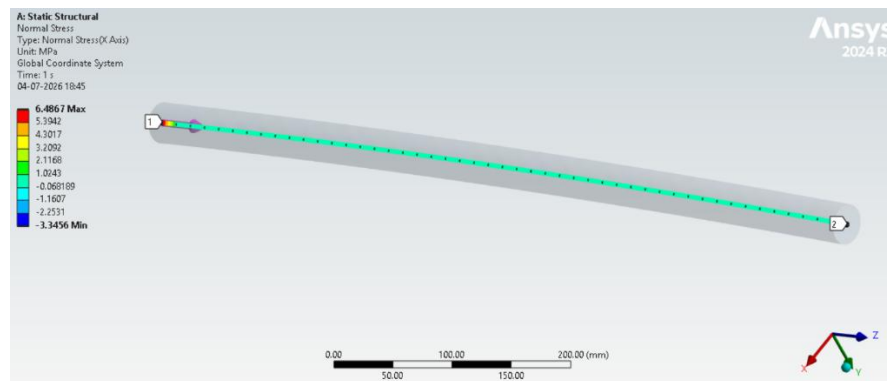
The stress distribution remains relatively uniform along most of the pillar length, with localized stress concentration observed near the constrained end due to the boundary conditions. This is expected because the fixed support restricts displacement, producing higher internal stresses in that region.

The obtained stress value remains significantly lower than the typical allowable strength of aerospace composite materials, indicating that the pillar operates within the elastic region under the applied loading. Therefore, the design is structurally safe for the specified load case.

Observation

- Maximum Equivalent Stress: 59.114 MPa
- Minimum Equivalent Stress: 8.676 MPa
- Highest stress occurs near the fixed support.
- No abnormal stress concentrations are observed along the pillar.
- The stress distribution indicates efficient load transfer through the structure.

Result : Normal Stress (X-Axis)



The Normal Stress (X-Axis) result represents the axial stress developed along the longitudinal direction of the composite pillar under the applied static loading. The analysis indicates that the maximum tensile normal stress is 6.487 MPa, while the minimum compressive normal stress is -3.346 MPa.

As shown in Figure 4.3, the highest tensile stress is concentrated near the fixed support, where the structural constraint generates greater internal resistance to the applied load. The stress gradually decreases along the length of the pillar, resulting in a nearly uniform stress distribution away from the support. The presence of both tensile and compressive stresses indicates the expected structural response under the applied loading and boundary conditions.

The obtained normal stress values are considerably lower than the allowable strength of typical aerospace composite materials, indicating that the pillar remains within the elastic region and is capable of safely sustaining the applied load without material failure.

Observations

- Maximum Normal Stress: 6.487 MPa (Tensile)
- Minimum Normal Stress: -3.346 MPa (Compressive)
- Maximum stress occurs near the fixed support due to load transfer and constraint effects.
- Stress distribution is smooth along the pillar length with no significant stress concentrations.
- The composite pillar exhibits stable load-carrying behavior and satisfies the structural requirements under the given loading condition.

Variation of Total Deformation Along the Length of the Composite Pillar

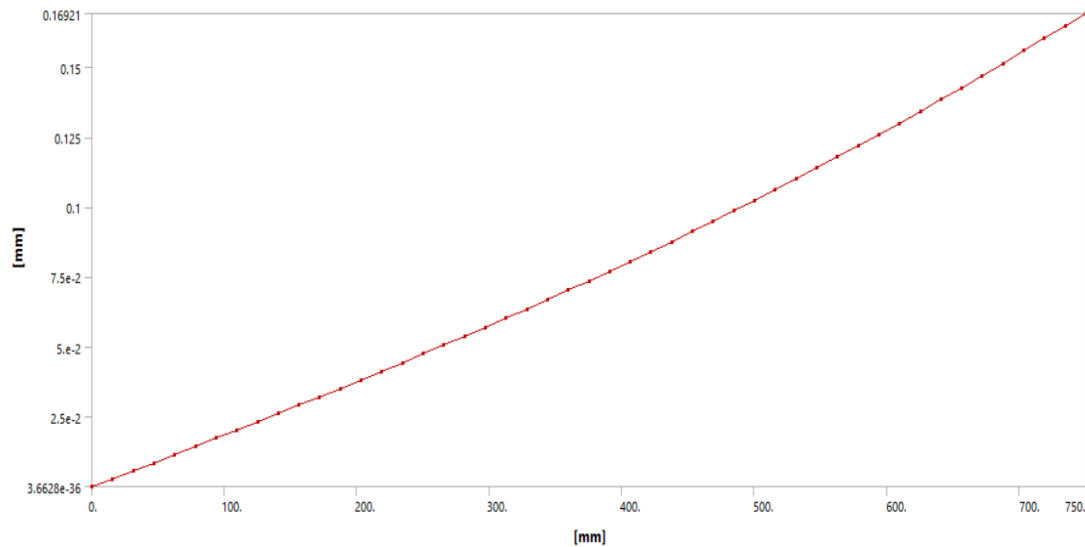
Figure, illustrates the variation of total deformation with respect to the pillar length. The deformation starts from approximately zero at the fixed support, satisfying the prescribed boundary condition, and gradually increases toward the free end.

The deformation curve exhibits a smooth and continuous increase throughout the length of the pillar, reaching a maximum value of approximately 0.169 mm at the free end. This trend is characteristic of a cantilever-type structural member subjected to static loading, where displacement accumulates progressively from the constrained end to the point of load application.

The absence of sudden changes or discontinuities in the deformation profile indicates good structural stiffness and uniform load distribution throughout the composite pillar. The relatively small maximum displacement further confirms that the structure possesses sufficient rigidity to withstand the applied load without excessive deflection.

Observations

- Minimum deformation: 0 mm at the fixed support.
- Maximum deformation: 0.169 mm at the free end.
- Deformation increases smoothly along the pillar length.
- The deformation profile confirms stable structural behavior and high stiffness of the composite pillar.



Variation of Normal Stress Along the Length of the Composite Pillar

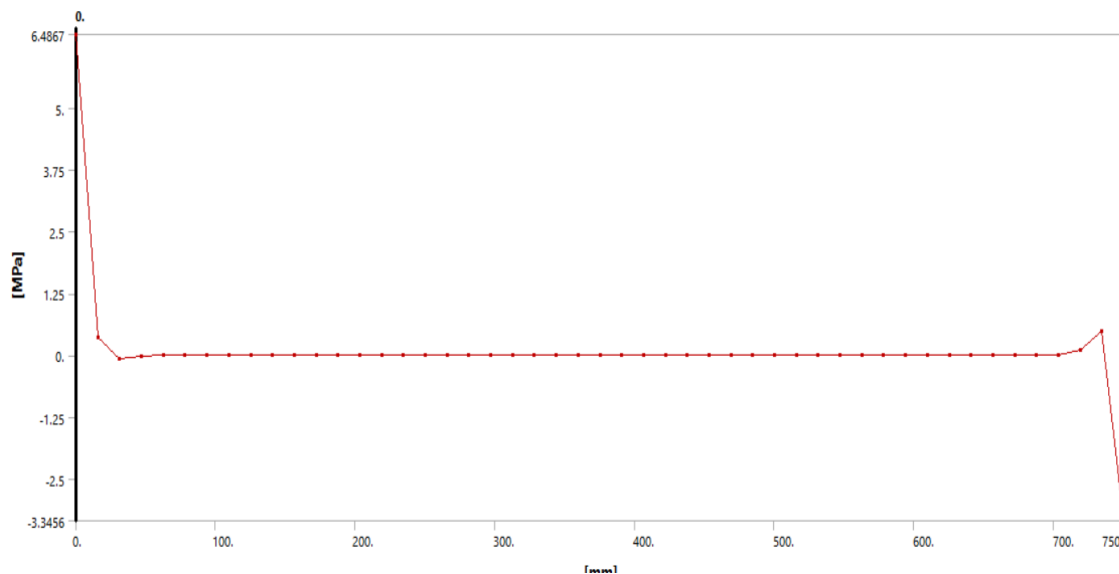
Figure presents the variation of normal stress along the length of the composite pillar. The graph indicates that the maximum tensile normal stress of 6.487 MPa occurs at the fixed end (0 mm), where the structural constraint produces the highest internal stresses. Immediately away from the support, the stress decreases rapidly and remains nearly constant, with values close to zero along most of the pillar length.

Near the free end, a slight increase in tensile stress is observed before the stress changes to a maximum compressive value of -3.346 MPa at the extreme end. This variation is mainly caused by the applied boundary conditions and the transfer of load through the structure.

The stress distribution demonstrates that the majority of the composite pillar experiences relatively low axial stress, while localized stresses are confined to the constrained and loaded regions. This behavior confirms efficient load transfer and indicates that no excessive stress concentrations exist along the structural member.

Observations

- Maximum tensile stress: 6.487 MPa at the fixed end.
- Stress rapidly decreases from the support and remains nearly uniform along the central portion of the pillar.
- Maximum compressive stress: -3.346 MPa at the free end.
- The stress profile indicates stable structural performance under the applied **loading**.



Conclusion :

The static structural analysis of the composite pillar was carried out using ANSYS Workbench to evaluate its structural performance under the applied loading conditions. The results showed a maximum total deformation of approximately 0.169 mm, indicating that the pillar possesses high stiffness and experiences only a small displacement under the applied load. The maximum equivalent (Von Mises) stress of 59.114 MPa and maximum normal stress of 6.487 MPa remained within the allowable limits for aerospace-grade composite materials, confirming that the structure operates safely within the elastic region.

The stress distribution revealed that the highest stresses were concentrated near the fixed support due to the boundary constraints, while the deformation increased gradually along the pillar length and reached its maximum at the free end. The smooth variation in both stress and deformation indicates effective load transfer and the absence of significant stress concentrations that could lead to premature structural failure.

Overall, the analysis demonstrates that the composite pillar provides an excellent combination of high strength, stiffness, and low deformation, making it suitable for lightweight aerospace structural applications. The finite element results validate the structural integrity of the design and indicate that the composite pillar can safely withstand the prescribed loading conditions while maintaining reliable mechanical performance.